Inorganic Cumulative Exam
February 12, 2005

1. $Mn_2(CO)_{10}$ can potentially exist in at least two forms that satisfy the 18-electron rule. $Cp_2Fe_2(CO)_4$ can also exist in multiple forms. IR spectroscopy has been a major tool in the study of this aspect of the chemistry. Give examples of the different geometric isomers for each complex and indicate how IR spectroscopy can distinquish between the options. Be very specific in your answer by choosing a geometry and indicating the expected IR bands.

2. A synthetic strategy commonly used with organometallic compounds is reduction with very active metals or metal hydrides. Indicate the products expected in the following reactions:

$Fe(CO)_5 + Na \rightarrow$

$Cr(CO)_6 + Na/Hg \rightarrow$

$Co_2(CO)_8 + Li[HBEt_3] \rightarrow$

$Mn_2(CO)_{10} + KH \rightarrow$

3. "Cp" is a very common ligand used in organometallic chemistry. The free ligand, however, is unstable and must be prepared immediately before use. Describe the preparation. Describe the synthesis of $Fe(Cp)_2$ starting with simple salts of Fe.

4. Cp can exhibit multiple binding modes. Starting with a Cp-containing complex, or the free ligand, suggest a route for the synthesis of three complexes which differ in the way Cp is bound to the metal ($FeCp_2$ cannot be used as an example). Indicate clearly, how the binding is different, and why you think the complexes should be sufficiently stable to isolate.